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Michael T. Saban

Computer Science PhD Student

Education

2024–Present **PhD in Computer Science**, *Loyola University Chicago*, Chicago, IL
Advisors: Dr. Samie Tootooni and Dr. Dmitriy Dligach. Focus: clinical NLP/LLMs, multimodal machine learning, and prehospital stroke triage.

2021–2023 **MS in Bioinformatics**, *Loyola University Chicago*, Chicago, IL
Completed via 4+1 Accelerated Program. GPA: 3.9/4.0.

2018–2022 **BS in Biology**, *Loyola University Chicago*, Chicago, IL
Honors; Cum Laude.

Research & Professional Experience

2024–Present **Graduate Research Assistant**, *Health Informatics & Data Science, LUC*, Chicago, IL

- **Multimodal Clinical Modeling:** Built an end-to-end multimodal fusion pipeline on MIMIC-IV (30-day post-discharge mortality), comparing early, intermediate, and late fusion of structured EHR data and clinical notes (TabPFN, XGBoost, BioClinical-ModernBERT; LoRA SFT of Llama-3; TIES / DARE-TIES adapter merging).
- **Clinical NLP / LLMs:** Developed a hybrid dense + BM25 retrieval system with an LLM-verification stage for schema-guided clinical information extraction of synthetic nursing dictations (MEDIQA-SYNUR 2026 shared task, LREC/ClinicalNLP).
- **Prehospital Stroke Triage:** Developed and validated ML models on EMS data to predict stroke and severe-stroke events; led the workflow / implementation study and clinician co-design.
- **Implementation Science:** Conducted qualitative analysis (CFIR) and customer discovery (I-Corps) to connect model development with real EMS workflows; collaborated with paramedics and clinicians on paramedic-centric interface design.

2023–2024 **Data Scientist**, *Health Informatics & Data Science, Parkinson School, LUC*, Chicago, IL
Aug 2023 – Aug 2024. Preceded by a Bioinformatics Internship (May–Aug 2023; MS capstone).

- Conducted a retrospective analysis of outpatient EHR data to evaluate hypertension management across a community health network.
- Investigated the association between Social Determinants of Health (Area Deprivation Index) and therapeutic inertia using statistical modeling.
- Disseminated findings via a moderated digital poster at AHA Scientific Sessions and a peer-reviewed publication in the *Journal of Human Hypertension* (Nature Portfolio).

Publications

Peer-Reviewed Journal Articles

2026 **Saban MT**, Hiura G, de la Peña P, Wozniak A, Heiferman D, Akbilgic O, Cichon M, Tootooni S. Machine learning models powered by emergency medical services data enhance stroke triage in prehospital settings. *Scientific Reports*. 2026;16(1):7139.

2025 **Saban MT**, Tootooni S, Markossian TW, Wozniak A, Hiura GT, Probst B, Habicht K, Kramer HJ. The association of area deprivation index and blood pressure control and therapeutic inertia among older adults with hypertension. *Journal of Human Hypertension*. 2025;39(11):748–754.

Peer-Reviewed Conference & Workshop Papers

2026 **Saban MT**, Yaghoubi A, Eslami B, Tootooni S, Dligach D. Lakefront AI Ramblers at MEDIQA-SYNUR 2026: Hybrid Retrieval and LLM Verification for Open-Source Schema-Guided Clinical Information Extraction. In: *Proceedings of the 8th Workshop on Clinical Natural Language Processing (Clinical NLP) @ LREC 2026*. 2026:212–221.

Published Abstracts & Proceedings

2025 **Saban MT**, Akbilgic O, Hiura G, de la Peña P, Heiferman D, Cichon M, Tootooni S. Machine learning analysis of EMS data leads to accurate and timely stroke and severe stroke detection. *Circulation*. 2025;152(Suppl_3):A4366491.

2025 **Saban MT**, de la Peña P, Heiferman D, Cichon M, Tootooni S. Evaluating machine learning models for prehospital stroke triage in emergency medical services. In: *AMIA Annual Symposium 2025*. 2025. Poster P49.

2024 **Saban MT**, Tootooni S, Markossian TW, Wozniak A, Hiura GT, Probst B, Habicht K, Kramer HJ. Neighborhood disadvantage is associated with uncontrolled blood pressure and therapeutic inertia among older adults with hypertension. *Circulation*. 2024;150(Suppl_1): A4141959.

Manuscripts Under Review

2026 **Saban MT**, Kasaie A, Olalekan P, Morrato EH, Tootooni S. Assessing the emergency medical services workflow for implementing an AI-driven tool for stroke triage. *Journal of Clinical and Translational Science*. Under review, minor revisions.

Manuscripts In Preparation

Saban MT, Tootooni S, Yoon W, Miller T, Afshar M, Dligach D. Multimodal fusion of structured EHR data and clinical notes for clinical outcome prediction. (First author).

Saban MT, Ahmad M, Dincer U, Kasaie A, Markossian T, Tootooni S. Modeling the economic impact of an AI-based clinical decision support tool for prehospital stroke triage. (First author).

Hossain A, **Saban MT**, Cichon M, Tootooni S. Paramedic perspectives on co-design of a prehospital stroke decision-support tool. (Co-author / mentor).

Dincer U, **Saban MT**, Kasaie A, Hossain A, Heiferman D, Markossian T, Tootooni S. Financial burden of stroke patients arriving at non-specialized hospitals. (Co-author / mentor).

Grants & Funded Research

Intramural

2025–2026 **StrokePath: An AI-Powered Decision Support Tool for Stroke Triage in Emergency Medical Services**, *PEP-STEP Award, Parkinson School, LUC*, $4,950
Student Researcher (PI: S. Tootooni). **Funded.**

2024–2025 **Artificial Intelligence as a Tool for Critical Stroke Prediction in Prehospital Settings**, *Cardiovascular Research Institute Micro-Grant, LUC*, $5,000
Student Researcher (PI: S. Tootooni). **Funded.**

Extramural

2024–2025 **Evaluating the Economic Impact of Stroke Misdiagnosis in Emergency Medical Services**, *I-Corps Seed Award, Institute for Translational Medicine, CTSA, NIH*, $2,500
Co-Investigator (PI: S. Tootooni). **Funded.**

2024–2025 **Scaling Up an Intelligent Stroke Detection Tool for EMS Across Chicagoland and Developing Strategic Implementation Plans**, *Implementation Seed Award, ITM 3.0 Dissemination & Implementation Cluster, CTSA, NIH*, $25,000
Co-Investigator (PI: S. Tootooni). Not funded.

Presentations

Oral Presentations

Dec 2025 **Economic Impact of EMS Stroke Mistriage Using HCUP-NEDS Data**, *INFORMS Regional Analytics Conference*, Chicago, IL
Co-author; presented by U. Dincer (mentee), who received the INFORMS Chicago Chapter 2025 Impactful OR/Analytics Prize, Student of the Year, for this project.

Oct 2024 **Assessing the Emergency Medical Services Workflow for Implementation of an AI-Driven Tool for Stroke Triage**, *INFORMS Annual Meeting*, Seattle, WA
First author; presented by A. Kasaie.

Posters

Nov 2026 **Multimodal Fusion of Structured EHR Data and Clinical Notes for 30-Day Mortality Prediction**, *AMIA Annual Symposium 2026*, Dallas, TX
Accepted.

Mar 2026 **Multimodal EHR Fusion (preliminary) & EMS Stroke Triage**, *Salon '26, Loyola University Chicago*, Chicago, IL
Presented two posters.

Nov 2025 **Evaluating Machine Learning Models for Prehospital Stroke Triage in Emergency Medical Services**, *AMIA Annual Symposium 2025*, Atlanta, GA
Poster P49. Also presented at St. Albert's Day, LUC, Nov 6, 2025.

Nov 2024 **Neighborhood Disadvantage is Associated with Uncontrolled Blood Pressure and Therapeutic Inertia among Older Adults with Hypertension**, *AHA Scientific Sessions 2024*, Chicago, IL
Moderated digital poster. Also presented at St. Albert's Day, LUC, Nov 7, 2024.

Sep 2024 **Applying the Consolidated Framework for Implementation Research to Artificial Intelligence Solutions in Healthcare Practice**, *NOBCChE 51st Annual Meeting*, Orlando, FL
Co-author; presented by P. Olalekan (mentee), awarded the Tier One Advancing Science Conference Grant.

Sep 2023 **Leveraging GitHub for Enhanced Collaboration and Efficiency in Research Labs: A Case Study in Health Informatics**, *NOBCChE 50th Annual Meeting*, New Orleans, LA
Co-author; presented by R. Brown (mentee), awarded the Tier One Advancing Science Conference Grant.

Teaching & Instructional Experience

Jul 2026 **Research Project Leader**, *Warrior-Scholar Project, LUC*, Chicago, IL
Sole instructor for a week-long programming project in a STEM boot camp for ∼10 student veterans and active-duty service members, July 13–17, 2026. Delivered ten contact hours (2 hrs/day, M–F) teaching Python for numerical modeling and visualization of the physics of motion.

2026 **Workshop Lead: LLM Chain-of-Thought Prompt Engineering**, *Salon '26, LUC*, Chicago, IL
Interactive workshop delivered to ∼70 attendees.

2025 **Workshop Lead: First Machine Learning with Python**, *Salon '25, LUC*, Chicago, IL
Introduced Python machine learning to interdisciplinary attendees.

2025 **Guest Speaker**, *Data Sciences Institute, LUC*, Chicago, IL
SDOH analysis (PolicyMap) and NCBI/PDB tools for high-school biology students.

Invited Talks & Seminars

2026 **Invited Speaker: Prompt Engineering for Nursing**, *MSN Immersion Day, Marcella Niehoff School of Nursing, LUC*, Maywood, IL
Invited talk on LLM bias, hallucination, and platform selection for MS nursing students.

2025 **Seminar Speaker**, *CS Department Fall Seminar Series, LUC*, Chicago, IL
Machine learning on EMS data for prehospital stroke prediction.

2025 **Invited Speaker**, *Center for Health Outcomes and Informatics Research (CHOIR), LUC*, Maywood, IL
The Path to Implementation of AI for Prehospital Stroke Triage.

Professional Research Training

2024 **Innovation Corps (I-Corps™) @NCATS, NIH**, *Loyola University Chicago*
Completed I-Corps™ customer-discovery training as the PhD-student member of the StrokePath team (S. Tootooni, PI; A. Kasaie, PhD, postdoc). Conducted stakeholder interviews with EMS personnel and health-system decision makers to refine strategies for prehospital stroke detection and integration.

2025 **Responsible Conduct of Research and Scholarship (UNIV 370)**, *Loyola University Chicago*

Honors & Awards

2024–Present **Graduate Research Assistantship**, *Loyola University Chicago*
Full tuition remission, healthcare coverage, and stipend.

2025 **Graduate School Internal Travel Award**, *Loyola University Chicago*, $750
Travel to AMIA Annual Symposium.

2024 **Graduate School Internal Travel Award**, *Loyola University Chicago*, $750
Travel to AHA Scientific Sessions.

Mentorship & Leadership

2023–Present **Research Mentor**, *Health Informatics & Data Science, LUC*

- **U. Dincer** (MS, Business Data Analytics): stroke cost-modeling using HCUP-NEDS; presented at the INFORMS Regional Analytics Conference and St. Albert's Day; awarded the INFORMS Chicago Chapter 2025 Impactful OR/Analytics Prize, Student of the Year.
- **A. Hossain** (medical student): qualitative paramedic interviews for the co-design of a prehospital stroke decision-support tool; first author on a manuscript in preparation.
- **P. Olalekan** (research intern, 2024): CFIR-based implementation analysis; poster at NOBCChE 2024 and co-author on a manuscript under review at *Journal of Clinical and Translational Science*.
- **R. Brown** (research intern, 2023): research-lab collaboration workflows; poster at NOBCChE 2023.

Service

Peer Review

2026 **Reviewer**, *ACL Rolling Review (ARR)*

2026 **Reviewer**, *ClinicalNLP Workshop at LREC 2026*

2025–2026 **Reviewer**, *AMIA Annual Symposium*

2025 **Reviewer**, *JMIR Medical Informatics*

Technical Skills

Programming	Python and R (proficient); SQL, Java, Scala
ML & NLP	PyTorch, scikit-learn, Hugging Face Transformers, LangChain; deep learning, transformers, foundation models (e.g., TabPFN), parameter-efficient fine-tuning (LoRA), retrieval-augmented generation, embedding/encoder models
LLMs & Compute	Open-source LLMs and clinical encoders deployed on local GPU clusters; Ollama, Slurm, CUDA, Linux command line, Git
Methods	Predictive modeling and clinical decision support, multimodal data fusion, clinical information extraction, qualitative analysis (CFIR), customer discovery (I-Corps)
Data	Clinical and health datasets: MIMIC-IV, HCUP, EMS/prehospital, EHR